Nexora Technologies Pvt. Ltd.

Data, Analytics, AI & Automation Handbook

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1. Purpose, Scope & Data Operating Philosophy

a. Purpose

Nexora Technologies Pvt. Ltd. uses data, analytics, artificial intelligence, machine learning, and workflow automation to improve customer outcomes, operational efficiency, product capabilities, and management decision-making.

This handbook establishes the enterprise framework for:

I. Data ownership and stewardship
II. Data architecture and lifecycle management
III. Data quality
IV. Analytics and reporting
V. Business intelligence
VI. Data engineering
VII. Artificial intelligence and machine learning
VIII. Generative AI
IX. Enterprise automation
X. AI governance
XI. Data privacy and security
XII. Model risk management
XIII. Data and AI operations
XIV. Measurement and continuous improvement

The objective is to ensure that data and AI systems are useful, trustworthy, secure, explainable where appropriate, and operated responsibly.

b. Scope

This handbook applies to:

I. Data generated by Nexora systems
II. Customer data processed by Nexora
III. Employee and corporate data used for business purposes
IV. Analytical datasets
V. Data warehouses and data lakes
VI. Business intelligence platforms
VII. Machine learning models
VIII. Generative AI applications
IX. AI agents
X. Automated workflows
XI. Data APIs
XII. Data pipelines
XIII. Third-party data sources
XIV. Contractors and partners operating data or AI solutions on behalf of Nexora

The requirements apply across Bengaluru, Pune, Hyderabad, London, remote work environments, customer environments, and approved cloud infrastructure.

c. Data and AI Philosophy

Nexora treats data as an enterprise asset and AI as a controlled capability rather than an independent technology category.

The operating philosophy is:

> ***Make data trustworthy. Make decisions measurable. Make AI useful. Automate responsibly. Protect information throughout the lifecycle.***

d. Business Outcomes

Data and AI initiatives should contribute to one or more measurable outcomes:

I. Revenue growth
II. Customer retention
III. Operational efficiency
IV. Cost reduction
V. Risk reduction
VI. Product improvement
VII. Employee productivity
VIII. Customer experience
IX. Faster decision-making
X. New digital capabilities

Projects without an identifiable business outcome should not proceed to production solely because the underlying technology is available.

2. Data & AI Organization

a. Organization Structure

The Data & AI organization works across the company and maintains close operating relationships with Product, Engineering, Cloud & Infrastructure, Cybersecurity, Customer Delivery, Finance, Sales, Marketing, and Corporate Strategy.

Core capabilities include:

I. Data Engineering
II. Data Platform Engineering
III. Analytics Engineering
IV. Business Intelligence
V. Data Science
VI. Machine Learning Engineering
VII. Generative AI Engineering
VIII. AI Product Engineering
IX. Automation Engineering
X. Data Governance
XI. Data Architecture
XII. Data Quality

b. Leadership Responsibilities

The Chief Data & AI Officer is responsible for:

I. Enterprise data strategy
II. Analytics strategy
III. AI strategy
IV. Data governance
V. AI governance coordination
VI. Data platform investment
VII. Data quality strategy
VIII. Data and AI workforce capability
IX. Enterprise AI risk management

c. Data Governance Council

The Data Governance Council meets monthly.

Core participants include representatives from:

Function	Primary Responsibility
Data & AI	Enterprise data strategy
Engineering	Application data
Cloud & Infrastructure	Platform architecture
Cybersecurity	Data security

Function	Primary Responsibility
Legal	Regulatory requirements
Privacy	Personal-data protection
Finance	Financial data
Customer Delivery	Customer data
Product	Product analytics
Corporate Strategy	Business performance

The council reviews:

I. Critical data assets
II. Data-quality trends
III. Data ownership
IV. Data-access issues
V. Major analytical definitions
VI. Data incidents
VII. Retention requirements
VIII. Data-platform priorities

d. AI Governance Committee

The AI Governance Committee meets monthly and reviews:

I. High-impact AI systems
II. Generative AI applications
III. AI-agent deployments
IV. Model risk
V. AI safety issues
VI. Privacy implications
VII. Security concerns
VIII. Model performance
IX. Human oversight
X. AI exceptions

e. Data Roles

Nexora distinguishes between:

I. Data Owner

Accountable for the business meaning, appropriate use, and access requirements of a data domain.

II. Data Steward

Responsible for day-to-day data quality, definitions, metadata, and issue coordination.

III. Data Custodian

Responsible for technical storage, access controls, infrastructure, and operational protection.

IV. Data Consumer

An individual or system authorized to use data for an approved purpose.

V. Model Owner

Accountable for a deployed analytical or machine learning model.

VI. AI Product Owner

Accountable for the business outcome and operating behavior of an AI-enabled product.

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3. Enterprise Data Architecture

a. Architecture Principles

Nexora's data architecture follows these principles:

I. Data should have an identifiable owner
II. Systems of record should be clearly defined
III. Data should be reusable where appropriate
IV. Data movement should be traceable
V. Sensitive data should receive stronger controls
VI. Data pipelines should be observable
VII. Analytical definitions should be standardized
VIII. Production data should not be copied unnecessarily
IX. Data products should have measurable quality
X. Architecture should support controlled scalability

b. Enterprise Data Layers

The enterprise data architecture commonly contains:

1. Source Systems
2. Ingestion Layer
3. Raw Data Layer
4. Standardized Data Layer
5. Curated Data Layer
6. Analytics Layer
7. Semantic Layer
8. Consumption Layer

I. Source Systems

Sources may include:

- Customer applications
- Product databases
- CRM platforms

- ERP systems
- Finance systems
- Nexora PeopleHub
- Nexora ServiceDesk
- Nexora Workspace
- Nexora Identity
- Product telemetry
- Marketing systems
- External data providers

II. Ingestion Layer

The ingestion layer collects data through:

I. APIs
II. Batch files
III. Event streams
IV. Database replication
V. Application connectors
VI. Scheduled exports

III. Raw Data Layer

Raw data is retained in its source-oriented form where required for traceability.

Raw datasets should include:

I. Source identifier
II. Ingestion timestamp
III. Source timestamp
IV. Pipeline identifier
V. Schema version
VI. Record lineage information

IV. Standardized Data Layer

Standardization may include:

I. Type conversion
II. Naming normalization
III. Timestamp normalization
IV. Currency normalization
V. Country-code standardization
VI. Duplicate handling
VII. Reference-data mapping

V. Curated Data Layer

Curated datasets are designed for repeatable business and analytical use.

Curated data should have:

I. Defined ownership
II. Documented business meaning
III. Quality checks
IV. Access controls
V. Refresh expectations
VI. Retention requirements

c. Analytical Data Architecture

Nexora uses cloud-based analytical infrastructure for enterprise reporting and advanced analytics.

A typical analytical architecture may include:

Layer	Example Responsibility
Source	Operational applications
Ingestion	Batch and streaming collection
Storage	Cloud data storage
Warehouse	Structured analytical processing
Transformation	SQL and data pipelines
Semantic	Business metrics and definitions
BI	Dashboards and reporting
AI/ML	Predictive and generative workloads

d. Data Architecture Review

Major architecture changes require review by the appropriate technical and data governance stakeholders.

Review areas include:

I. Data classification
II. Data residency
III. Security
IV. Privacy
V. Scalability
VI. Cost
VII. Data lineage
VIII. Disaster recovery
IX. Operational ownership

4. Data Domains, Ownership & Metadata

a. Enterprise Data Domains

Nexora organizes important information into logical domains.

Examples include:

I. Customer
II. Product
III. Sales
IV. Finance
V. People
VI. Delivery
VII. Technology
VIII. Security
IX. Marketing
X. Operations

b. Critical Data Assets

A critical data asset is information whose loss, corruption, unavailability, or misuse could materially affect Nexora.

Examples include:

I. Financial reporting datasets
II. Customer contract information
III. Production product telemetry
IV. Employee master data
V. Security-event data
VI. Customer analytical datasets

Critical data assets require documented ownership.

c. Data Catalog

Nexora KnowledgeHub and approved data-catalog capabilities may contain metadata for important datasets.

Metadata should include:

I. Dataset name
II. Owner
III. Steward
IV. Description
V. Classification
VI. Source
VII. Refresh frequency
VIII. Retention period
IX. Quality expectations
X. Downstream dependencies

d. Business Glossary

The enterprise glossary defines important terms.

Examples include:

Customer

An organization or business entity with an active commercial relationship with Nexora.

Revenue

Recognized revenue according to Finance-approved accounting definitions.

Active Customer

A customer meeting the approved operational definition for the relevant reporting period.

Production Incident

A service event meeting the incident classification criteria established by Engineering and Operations.

Definitions must not be changed independently inside individual dashboards when an enterprise definition already exists.

e. Metric Ownership

Every enterprise KPI should have:

I. Metric name
II. Definition
III. Calculation logic
IV. Owner
V. Source system
VI. Refresh frequency
VII. Approved consumers
VIII. Effective date

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5. Data Lifecycle Management

a. Data Lifecycle

Data generally passes through:

I. Creation
II. Collection
III. Ingestion
IV. Storage
V. Transformation
VI. Use
VII. Sharing
VIII. Archival
IX. Deletion

Controls must be considered at each stage.

b. Data Creation

Data producers are responsible for capturing information accurately.

Applications should avoid collecting unnecessary fields.

Required validation should occur as close to the source as practical.

c. Data Storage

Storage systems must match:

I. Data classification
II. Availability requirements
III. Retention period
IV. Performance requirements
V. Access requirements
VI. Regulatory obligations

d. Data Sharing

Data may be shared only when:

I. The recipient is authorized
II. The purpose is legitimate
III. The information is appropriate for the purpose
IV. Security controls are adequate
V. Contractual requirements are satisfied

e. Data Archival

Archival may be used where information must be retained but is not regularly accessed.

Archived information should remain:

I. Protected
II. Discoverable when legitimately required
III. Subject to retention controls
IV. Subject to deletion requirements

f. Data Deletion

Data should be securely deleted when:

I. Retention expires
II. Legal requirements permit deletion
III. The business purpose ends
IV. Customer requirements require deletion

Deletion must consider primary stores, derived datasets, exports, and applicable backups.

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6. Data Quality Management

a. Data Quality Principles

Data quality is evaluated using multiple dimensions.

Dimension	Meaning
Accuracy	Data represents the intended real-world value
Completeness	Required information is present
Consistency	Values agree across systems
Timeliness	Data is available when required
Validity	Data conforms to defined rules
Uniqueness	Duplicate records are controlled

b. Data Quality Targets

Critical analytical datasets should establish measurable thresholds.

Example standards:

Quality Dimension	Example Target
Completeness	≥98%
Validity	≥99%
Duplicate rate	<1%
Pipeline success	≥99%
Timely refresh	≥98%
Critical-field accuracy	≥99%

Targets may vary by dataset risk and business purpose.

c. Data Quality Rules

Examples include:

I. Customer ID cannot be null
II. Transaction amount cannot be negative unless explicitly allowed
III. Currency must use an approved ISO code
IV. Country must use an approved reference value
V. Event timestamp must fall within an accepted range
VI. Primary keys must be unique

d. Data Quality Incidents

A quality issue should be logged when data defects materially affect:

I. Customer reporting
II. Financial reporting
III. Executive dashboards

IV. Production applications
V. AI or ML models
VI. Regulatory reporting

e. Data Quality Root Cause Analysis

Repeated data-quality failures require analysis of:

I. Source application
II. Pipeline transformation
III. Business rule
IV. Reference data
V. Human process
VI. Schema change
VII. Integration failure

The preferred solution is to correct the source or process rather than repeatedly patching downstream outputs.

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7. Data Engineering & Pipelines

a. Pipeline Engineering Principles

Production data pipelines should be:

I. Repeatable
II. Observable
III. Recoverable
IV. Version controlled
V. Testable
VI. Documented
VII. Secure

b. Batch Processing

Batch pipelines may be used for:

I. Daily reporting
II. Financial reconciliation
III. Historical data processing
IV. Periodic customer data ingestion
V. Scheduled model training

Batch schedules should account for upstream dependencies.

c. Streaming Data

Streaming may be used where near-real-time processing provides measurable value.

Examples include:

I. Product telemetry
II. Security events
III. Operational monitoring
IV. Customer activity
V. Event-driven automation

Streaming should not be introduced solely because it is technically available.

d. Pipeline Reliability

Critical pipelines should define:

I. Owner
II. Schedule
III. SLA
IV. Expected volume
V. Failure behavior
VI. Retry behavior
VII. Alerting
VIII. Recovery procedure

e. Schema Changes

Source-system schema changes must be communicated to downstream owners when they could affect data pipelines.

Material changes may require:

I. Compatibility testing
II. Pipeline updates
III. Dashboard validation
IV. Model impact analysis
V. Documentation updates

f. Pipeline Monitoring

Monitoring should include:

I. Execution status
II. Duration
III. Record counts
IV. Error counts
V. Freshness
VI. Schema changes
VII. Data-quality failures

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8. Analytics, Business Intelligence & Reporting

a. Analytics Categories

Nexora uses four primary analytical categories.

I. Descriptive Analytics

Answers:

What happened?

Examples:

- Revenue reporting
- Customer usage
- Delivery performance
- Employee metrics

II. Diagnostic Analytics

Answers:

Why did it happen?

Examples:

- Revenue variance
- Customer churn drivers
- Delivery delays
- Conversion changes

III. Predictive Analytics

Answers:

What is likely to happen?

Examples:

- Demand forecasting
- Customer churn prediction
- Capacity forecasting
- Revenue forecasting

IV. Prescriptive Analytics

Answers:

What should we do?

Examples:

- Resource allocation
- Pricing recommendations
- Customer prioritization
- Operational optimization

b. Dashboard Standards

Executive dashboards should prioritize:

I. Decision-relevant metrics
II. Clear definitions
III. Consistent time periods
IV. Appropriate comparisons
V. Visible data freshness
VI. Exception identification

Dashboards should not use visual complexity to compensate for weak analysis.

c. Dashboard Ownership

Every important dashboard should have:

I. Business owner
II. Technical owner
III. Data sources
IV. Refresh schedule
V. Metric definitions
VI. Last validation date

d. Executive Reporting

Executive reports should distinguish between:

I. Actual values
II. Forecast values
III. Targets
IV. Variance
V. Confidence or uncertainty where relevant

e. Self-Service Analytics

Self-service analytics is encouraged when:

I. Users have approved data access
II. Definitions are documented
III. Sensitive data is protected
IV. Outputs are not represented as official reporting without validation

f. Official Reporting

Reports used for:

I. Financial reporting
II. Regulatory submissions
III. Executive performance management
IV. Customer contractual reporting

must use approved definitions and controlled data sources.

9. Data Science & Machine Learning

a. Machine Learning Lifecycle

Nexora's standard ML lifecycle is:

1. Business problem definition
2. Data assessment
3. Feature engineering
4. Model development
5. Validation
6. Risk review
7. Deployment
8. Monitoring
9. Retraining
10. Retirement

b. Problem Definition

Every ML initiative must establish:

I. Business objective
II. Target variable
III. Prediction horizon
IV. Success metric
V. Baseline
VI. Operational decision affected

c. Dataset Preparation

Data scientists must evaluate:

I. Missing values
II. Duplicates
III. Outliers
IV. Data leakage
V. Class imbalance
VI. Sampling bias
VII. Label quality
VIII. Temporal consistency

d. Model Evaluation

Evaluation metrics should match the business problem.

Examples:

Problem	Possible Metrics
Classification	Precision, recall, F1, ROC-AUC
Regression	MAE, RMSE, R^2
Forecasting	MAE, RMSE, MAPE
Ranking	NDCG, MAP
Clustering	Silhouette score

A high statistical score does not automatically establish business usefulness.

e. Baselines

Every predictive model should be compared against an appropriate baseline.

Possible baselines include:

I. Historical average
II. Previous-period value
III. Simple rule
IV. Existing production model
V. Human decision process

f. Model Documentation

Production models should document:

I. Purpose
II. Training data
III. Features
IV. Model type
V. Evaluation results
VI. Limitations
VII. Owner
VIII. Deployment date
IX. Monitoring metrics
X. Retraining conditions

10. Machine Learning Operations & Model Governance

a. Model Registry

Production models should have identifiable records.

A model record may include:

Model ID: NXR-ML-2026-087

Owner: Data Science

Business Function: Customer demand forecasting

Model Type: Gradient boosting

Training Dataset: CustomerDemand_v12

Production Date: 12 August 2026

Review Cycle: Quarterly

b. Model Versioning

Model versions should be traceable to:

I. Code version
II. Training dataset
III. Feature definitions
IV. Hyperparameters
V. Evaluation results
VI. Deployment version

c. Model Monitoring

Production models should be monitored for:

I. Performance degradation
II. Data drift
III. Feature drift
IV. Prediction distribution changes
V. Availability
VI. Latency
VII. Business outcome changes

d. Model Drift

Drift may occur when the relationship between input data and outcomes changes.

Triggers for investigation may include:

I. Significant input-distribution changes
II. Accuracy decline
III. Increased error rates
IV. Customer behavior changes
V. Product changes
VI. Market changes

e. Retraining

Retraining may be triggered by:

I. Scheduled review
II. Performance degradation
III. Material data changes
IV. Business-process changes
V. New training data
VI. Model drift

Retraining does not automatically mean immediate deployment.

f. Model Retirement

Models should be retired when:

I. Business purpose ends
II. Performance remains below threshold
III. A superior model replaces it
IV. Data source becomes unavailable
V. Risk exceeds acceptable levels

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11. Artificial Intelligence & Generative AI

a. AI Principles

Nexora applies the following AI principles:

I. Business usefulness
II. Human accountability
III. Security
IV. Privacy
V. Transparency
VI. Reliability
VII. Appropriate human oversight
VIII. Risk-based governance

b. AI Use Cases

Approved AI use cases may include:

I. Document summarization
II. Enterprise search
III. Customer-support assistance
IV. Software development assistance
V. Analytics assistance
VI. Classification
VII. Forecasting
VIII. Workflow automation
IX. Knowledge extraction
X. Internal productivity

c. Generative AI

Generative AI systems may generate:

I. Text
II. Code
III. Summaries
IV. Structured outputs
V. Classifications
VI. Recommendations
VII. Analytical explanations

Outputs must be reviewed according to the risk of the use case.

d. Approved AI Platforms

Nexora may use approved enterprise AI platforms such as Nexora Assist and authorized third-party AI services.

Employees must not submit Confidential or Restricted information to unapproved public AI services.

e. Prompt and Context Management

AI applications must control:

I. System instructions
II. User permissions
III. Retrieved context
IV. Sensitive information
V. Tool permissions
VI. Output handling

AI systems must not receive broader data access than required for the requested task.

f. Hallucination Risk

AI-generated information may be incorrect.

For high-impact use cases, outputs must be validated against authoritative sources.

Examples include:

I. Financial decisions
II. Legal interpretations
III. Security decisions
IV. Customer contractual commitments
V. Regulatory submissions
VI. Employee decisions

g. Human Oversight

Human review is required when an AI output could materially affect:

I. Legal rights
II. Financial outcomes
III. Employment decisions
IV. Security actions
V. Customer contractual obligations
VI. Regulatory obligations

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12. AI Agents, RAG & Intelligent Automation

a. AI Agent Definition

An AI agent is an AI-enabled system capable of interpreting objectives and taking one or more actions through approved tools, APIs, or enterprise systems.

Agents may:

I. Retrieve information
II. Call APIs
III. Execute workflows
IV. Analyze data
V. Generate documents
VI. Trigger notifications
VII. Recommend actions

b. Agent Architecture

A typical Nexora agent may contain:

1. User Interface
2. Identity Layer
3. Agent Orchestrator
4. Model
5. Retrieval Layer
6. Tool Layer
7. Policy Layer
8. Logging Layer
9. Human Approval Layer

c. Tool Permissions

Agent tools must follow least privilege.

An agent that can read a system should not automatically receive write access.

An agent capable of sending communications should have explicit authorization to perform that action.

d. High-Risk Agent Actions

Additional controls are required for agents capable of:

I. Financial transactions
II. Account creation
III. Access changes
IV. Production configuration changes
V. Customer communications
VI. Data deletion
VII. Security-control modification

e. Retrieval-Augmented Generation

RAG systems may retrieve enterprise information from approved sources before generating responses.

RAG implementations should control:

I. Source authorization
II. Document classification
III. Retrieval scope
IV. Prompt construction
V. Citation or provenance where appropriate
VI. Prompt injection risks
VII. Output validation

f. Retrieval Quality

RAG systems should be evaluated for:

I. Retrieval relevance
II. Context completeness
III. Groundedness
IV. Answer accuracy
V. Unauthorized data exposure

g. Agent Evaluation

Example minimum targets for a production internal AI agent may include:

Metric	Target
Task success	≥90%
Grounded responses	≥95%
Critical safety failures	0
Median response time	≤5 seconds
Unauthorized tool execution	0

Targets may be increased for higher-risk applications.

13. AI Safety, Security, Privacy & Responsible Use

a. AI Security Risks

AI systems must consider:

I. Prompt injection
II. Data leakage
III. Insecure tool use
IV. Excessive permissions
V. Model manipulation
VI. Training-data risks
VII. Malicious input
VIII. Output-based attacks

b. Prompt Injection

AI applications that retrieve external or enterprise content must assume that retrieved content may contain instructions intended to manipulate the model.

Systems should separate:

I. Trusted system instructions
II. User instructions
III. Retrieved information
IV. Tool outputs

Retrieved content must not automatically gain authority over the agent.

c. Sensitive Information

AI systems must apply data-classification controls.

Restricted information should only be processed by approved AI systems with appropriate controls.

d. AI Privacy Assessment

A privacy review may be required when AI systems process:

I. Personal information
II. Employee data
III. Customer information
IV. Behavioral data
V. Sensitive personal information

e. Model Explainability

Where decisions materially affect individuals or customers, the organization should be able to explain:

I. What the system does
II. What information it uses
III. What factors influence the result

IV. What limitations exist
V. How a human can review the result

f. Bias and Fairness

Relevant AI systems should be evaluated for inappropriate differences in outcomes across meaningful groups.

The assessment should consider:

I. Dataset composition
II. Label quality
III. Feature selection
IV. Error rates
V. Business impact

Fairness testing must be proportionate to the use case.

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14. Enterprise Automation & Workflow Governance

a. Automation Philosophy

Automation should remove repetitive work while preserving accountability.

A workflow should be automated when:

I. The process is sufficiently understood
II. Inputs are predictable
III. Failure modes are manageable
IV. Business ownership is clear
V. Expected value exceeds implementation and operating cost

b. Nexora Flow

Nexora Flow is the enterprise workflow automation platform.

Common automation categories include:

I. Notifications
II. Data synchronization
III. Approval workflows
IV. Report generation
V. Service requests
VI. Scheduled tasks
VII. Operational checks
VIII. Document processing

c. Automation Ownership

Every production automation must have:

I. Business owner
II. Technical owner
III. Trigger definition
IV. Expected output
V. Failure procedure
VI. Monitoring
VII. Access permissions

d. Automation Risk Levels

Level	Example	Governance
Low	Internal notification	Standard
Moderate	Data transformation	Technical review
High	Customer-facing workflow	Formal review
Critical	Financial or security action	Human approval

e. Automation Failure

Failed workflows should generate sufficient information for investigation.

Minimum information may include:

I. Workflow ID
II. Execution timestamp
III. Trigger
IV. Failed step
V. Error category
VI. Retry status
VII. Owner

f. Human Approval

Human approval should remain in workflows where automation could cause material irreversible impact.

Examples include:

I. Financial transfers
II. Contract execution
III. Customer termination
IV. Destructive production changes
V. Deletion of critical records

15. Data & AI Security, Privacy & Access Controls

a. Data Access

Access to analytical datasets must follow least privilege.

Access decisions consider:

I. User identity
II. Role
III. Business purpose
IV. Data classification
V. Customer obligations
VI. Geographic requirements

b. Analytical Workspace Security

Analytical environments should implement:

I. Individual identities
II. MFA
III. Role-based access
IV. Audit logging
V. Controlled exports
VI. Encryption

c. Production Data

Production datasets must not be copied into uncontrolled development environments.

Where development requires representative data, teams should prefer:

I. Synthetic data
II. Masked data
III. Anonymized data
IV. Approved limited extracts

d. Data Exports

Exports containing Confidential or Restricted information must be controlled.

Users must consider:

I. Destination
II. Recipient
III. Encryption
IV. Retention
V. Deletion

e. Model Access

Production model endpoints should use authenticated access.

Model APIs should enforce:

I. Authentication
II. Authorization
III. Rate limits
IV. Logging
V. Input validation

f. AI Tool Access

AI agents and applications must receive only the data and tools required for their intended function.

A customer-support agent should not receive unrestricted access to finance or People Operations datasets merely because those systems are technically reachable.

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16. Data & AI Operations, Reliability & Cost Management

a. Data Service Levels

Critical data products should define service expectations.

Examples include:

Data Product	Refresh Target	Availability
Executive KPI dataset	Daily	≥99%
Customer analytics dataset	Daily	≥99.5%
Operational event stream	Near real time	≥99.9%
Finance reporting dataset	Monthly/controlled	≥99.9%

b. Data Pipeline Incidents

A data incident may be declared when:

I. Critical pipeline fails
II. Data becomes materially incorrect
III. Required refresh is missed
IV. Sensitive information is exposed
V. Downstream business reporting is affected

c. Data Recovery

Critical pipelines should have recovery procedures covering:

I. Source availability
II. Failed transformations
III. Partial loads
IV. Duplicate records
V. Corrupt datasets
VI. Historical backfills

d. Data Cost Management

Data platforms are subject to cost controls.

Teams should monitor:

I. Storage growth
II. Query consumption
III. Compute usage
IV. Data-transfer costs
V. Unused datasets
VI. Duplicate pipelines

e. AI Cost Management

AI applications must monitor:

I. Model usage
II. Token consumption
III. Inference cost
IV. Tool execution cost
V. Retrieval cost
VI. Storage cost

High-volume AI applications should establish budgets and usage thresholds.

f. Cost Alerts

For material AI or data workloads, budget alerts may be configured at:

50%, 75%, 90%, and 100% of approved monthly budget.

Unexpected usage should be investigated rather than simply accepted as normal growth.

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17. Data & AI Project Lifecycle

a. Stage 1 — Opportunity Identification

The sponsoring business team documents:

I. Business problem
II. Current process
III. Expected value

IV. Users affected
V. Data required
VI. Proposed solution

b. Stage 2 — Feasibility

The team evaluates:

I. Data availability
II. Data quality
III. Technical feasibility
IV. Privacy
V. Security
VI. Expected cost
VII. Delivery timeline

c. Stage 3 — Prototype

Prototype work should be isolated from production systems.

Prototype success criteria should be measurable.

d. Stage 4 — Validation

Validation may include:

I. Functional testing
II. Data-quality testing
III. Model evaluation
IV. Security testing
V. Privacy review
VI. User acceptance

e. Stage 5 — Production Approval

Production approval requires:

I. Named owner
II. Monitoring
III. Incident procedure
IV. Security controls
V. Documentation
VI. Cost expectations
VII. Data-retention requirements

f. Stage 6 — Production Operation

Production systems are monitored continuously according to risk.

Owners review:

I. Performance
II. Reliability
III. Usage
IV. Cost
V. Quality
VI. Risk

g. Stage 7 — Retirement

Data and AI systems should be retired when:

I. Business value ends
II. Better solutions replace them
III. Costs become unjustified
IV. Risks become unacceptable
V. Source data becomes unavailable

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18. Data & AI Governance Metrics and Operating Review

a. Enterprise Data Metrics

The Data & AI organization reviews:

I. Data-quality scores
II. Pipeline success
III. Dataset freshness
IV. Data incidents
V. Access-review completion
VI. Critical dataset coverage
VII. Metadata completeness

b. Analytics Metrics

Analytics teams may track:

I. Dashboard adoption
II. Report usage
III. Query performance
IV. Decision-cycle time
V. Reporting defects
VI. Self-service adoption

c. AI Metrics

AI systems may be evaluated on:

I. Accuracy
II. Groundedness
III. Task success
IV. Latency
V. Cost per interaction
VI. Safety incidents
VII. User satisfaction
VIII. Escalation rate

d. Automation Metrics

Automation teams may track:

I. Hours saved
II. Execution success rate
III. Failure rate
IV. Manual interventions
V. Cost avoided
VI. Processing time
VII. Business adoption

e. Value Measurement

Every material data or AI initiative should establish a value hypothesis.

Example:

Current process: 5 hours per weekly reporting cycle

Automated process: 45 minutes

Expected saving: 4.25 hours per cycle

Annual cycles: 52

Estimated annual capacity recovered: 221 hours

Value calculations should use Finance-approved assumptions when incorporated into formal business cases.

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19. Reference Implementation: Customer Analytics Platform

a. Business Context

Nexora is implementing a customer analytics platform to consolidate customer engagement information from multiple systems.

Project ID: NXR-PRD-0427

The platform supports:

I. Customer engagement analysis
II. Usage reporting
III. Retention analysis
IV. Revenue analysis
V. Operational forecasting

b. Data Sources

The platform receives information from:

I. Product telemetry
II. CRM
III. Customer support systems
IV. Billing systems
V. Marketing platforms

c. Processing

Data passes through:

1. Source ingestion
2. Raw storage
3. Standardization
4. Customer identity resolution
5. Curated datasets
6. KPI computation
7. Dashboard consumption
8. Predictive analytics

d. Security

The platform uses:

I. Nexora Identity
II. Role-based access
III. Encryption
IV. Audit logging
V. Restricted production access

e. Analytics

Key measures include:

I. Active customers
II. Product usage
III. Feature adoption
IV. Support volume
V. Customer retention
VI. Revenue contribution

f. Forecasting

Forecasting models may estimate:

I. Customer demand
II. Usage trends
III. Revenue
IV. Support requirements

Models are monitored against actual outcomes.

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20. Reference Implementation: AI-Powered Enterprise Assistant

a. Business Objective

Nexora is developing AI capabilities through Nexora Assist to help employees retrieve approved enterprise information.

The assistant may answer questions about:

I. Corporate policies
II. IT procedures
III. Engineering standards
IV. Security requirements
V. Data standards
VI. People processes

b. Retrieval Architecture

A controlled retrieval architecture may include:

1. User authentication
2. Permission evaluation
3. Query processing
4. Document retrieval
5. Context filtering
6. Model generation
7. Response validation
8. Audit logging

c. Authorization

The assistant must not retrieve documents the user is not authorized to access.

Document-level authorization is preferred where practical.

d. Grounding

The assistant should prioritize authoritative enterprise information.

If sufficient information is unavailable, the system should state that it cannot establish the answer rather than inventing a policy.

e. Evaluation

The initial production evaluation target is:

Metric	Target
Task success	≥90%
Grounded responses	≥95%
Critical security failures	0
Unauthorized retrieval	0
Median response time	≤5 seconds

f. Human Escalation

Questions involving:

I. Legal interpretation
II. Employee disciplinary matters
III. Security incidents
IV. Financial approval
V. Contract commitments

should be routed to appropriate human functions rather than treated as authoritative decisions by the assistant.

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21. Reference Implementation: Automated Analytics Workflow

a. Workflow

A recurring analytics workflow may be implemented through Nexora Flow.

Example sequence:

1. Scheduler starts workflow.
2. Analytics data is refreshed.
3. Data-quality checks execute.
4. KPI calculations run.
5. Forecasting model executes.
6. Anomaly detection identifies unusual values.

7. AI validation reviews explanatory context.
8. Report is generated.
9. Stakeholders receive approved output.
10. Execution metadata is recorded.

b. Failure Handling

If the data-quality check fails:

I. Downstream model execution stops
II. Report generation is blocked
III. Workflow owner receives an alert
IV. Failure details are logged
V. Retry occurs according to policy

c. Business Benefit

A successfully automated process may reduce a recurring analyst workflow from several hours to less than one hour.

The business case must be validated against actual operating results.

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22. Data & AI Exception Management

a. Exception Requirement

An exception is required when a data or AI system cannot meet a mandatory enterprise requirement.

Examples include:

I. Temporary use of non-standard data storage
II. Delayed model validation
III. Temporary use of legacy integration
IV. Unavailable metadata requirement
V. Extended model review period

b. Exception Record

Each exception should include:

I. Exception ID
II. Requirement
III. Reason
IV. Affected system
V. Data classification
VI. Risk
VII. Compensating controls
VIII. Owner
IX. Expiration date

c. AI Exceptions

AI exceptions require additional consideration of:

I. Model behavior
II. User impact
III. Privacy
IV. Security
V. Human oversight
VI. Explainability

d. Expiration

Exceptions are temporary and must be reviewed before expiration.

An expired exception does not create permanent authorization.

Appendix A. Data Classification Quick Reference

Data Type	Typical Classification
Public product information	Public
Internal dashboard	Internal
Non-public customer metrics	Confidential
Customer contract data	Confidential
Employee master data	Confidential
Authentication credentials	Restricted
Encryption keys	Restricted
Sensitive personal data	Restricted
Security investigation data	Restricted

Appendix B. Data Quality Reference

Metric	Standard Target
Critical-field completeness	≥98%
Critical-field validity	≥99%
Critical dataset pipeline success	≥99%
Duplicate rate	<1%

Metric	Standard Target
Timely refresh	≥98%
Critical analytical dataset availability	≥99%

Targets may be stricter where contractual, regulatory, or business requirements apply.

Appendix C. Model Governance Reference

A production model record should contain:

- Model ID
- Model name
- Business owner
- Technical owner
- Business purpose
- Training data
- Feature set
- Model type
- Evaluation metrics
- Baseline
- Known limitations
- Risk classification
- Deployment date
- Version
- Monitoring metrics
- Retraining criteria
- Retirement criteria

Appendix D. AI Risk Classification

Risk Level	Example	Governance
Low	Internal text summarization	Standard
Moderate	Analytics recommendation	Review
High	Customer-facing AI recommendation	Formal AI review
Critical	AI influencing financial, legal, employment, or security decisions	Enhanced governance and human oversight

Risk classification must consider the actual use rather than only the underlying model.

Appendix E. AI Deployment Checklist

Before production deployment:

- [ ] Business owner assigned
- [ ] Technical owner assigned
- [ ] Data sources documented
- [ ] Data classification completed
- [ ] Privacy assessment completed where required
- [ ] Security review completed
- [ ] Model evaluation completed
- [ ] Baseline comparison completed
- [ ] Failure modes documented
- [ ] Human oversight defined
- [ ] Access controls configured
- [ ] Logging enabled
- [ ] Monitoring configured
- [ ] Cost limits established
- [ ] Incident procedure documented
- [ ] Rollback procedure documented
- [ ] Governance approval obtained where required

Appendix F. Data Pipeline Production Checklist

- [ ] Source identified
- [ ] Data owner assigned
- [ ] Pipeline owner assigned
- [ ] Schema documented
- [ ] Data classification completed
- [ ] Transformation logic version controlled
- [ ] Data-quality checks implemented
- [ ] Freshness monitoring enabled
- [ ] Failure alerts configured
- [ ] Retry behavior documented
- [ ] Backfill procedure documented
- [ ] Access controls configured
- [ ] Retention period established
- [ ] Cost monitored
- [ ] Disaster-recovery requirements evaluated

Appendix G. Data & AI Record Identifiers

Nexora uses structured identifiers for important data and AI records.

Examples include:

Identifier	Purpose
NXR-DATA-005	Data, Analytics, AI & Automation Handbook
NXR-DATASET-2026-014	Enterprise dataset
NXR-PIPE-2026-087	Data pipeline
NXR-ML-2026-087	Machine learning model
NXR-AI-2026-031	AI application
NXR-AGENT-2026-012	AI agent
NXR-AUTO-2026-044	Enterprise automation
NXR-DQ-2026-019	Data-quality issue
NXR-AI-EXC-2026-006	AI governance exception

Identifiers identify records and do not independently determine information classification.

Appendix H. Data Governance Calendar

Activity	Frequency
Data pipeline review	Weekly
Critical data-quality review	Weekly
Data Governance Council	Monthly
AI Governance Committee	Monthly
Data access review	Quarterly
Critical model review	Quarterly
Data architecture review	As required
AI risk assessment	Before material deployment
Data-retention review	Annual
Enterprise data strategy review	Annual

Appendix I. Roles and Responsibilities Summary

Function	Primary Data/AI Responsibility
Executive Leadership	Strategy and investment
Data & AI	Enterprise data and AI capabilities
Engineering	Application data and software integration
Cloud & Infrastructure	Platform and infrastructure
Cybersecurity	Security controls
Legal	Regulatory and contractual obligations
Privacy	Personal-data protection
Finance	Financial data and value measurement
Product	Product analytics and AI capabilities
Customer Delivery	Customer data and delivery outcomes
People Operations	Employee data
Procurement	Supplier data and third-party controls
All Employees	Responsible data use

Appendix J. Related Nexora Standards

Data, analytics, AI, and automation requirements should be interpreted together with applicable Nexora standards.

Document	Relationship
NXR-CORP-001	Company, People & Corporate Governance
NXR-TECH-002	Product, Engineering & Software Development
NXR-INFRA-003	Cloud, Infrastructure & IT
NXR-SEC-004	Cybersecurity, Risk, Privacy & Compliance
NXR-DEL-006	Customer Delivery & Customer Success
NXR-COM-007	Sales, Marketing, Finance & Procurement
NXR-OPS-008	Legal, Operations, Continuity & Knowledge

Where security, privacy, legal, contractual, or regulatory requirements are stricter than a general data or AI practice, the stricter applicable requirement must be followed unless an authorized exception exists.

Appendix K. Enterprise Data & AI Principles

Nexora's enterprise data and AI practices are governed by the following principles:

1. Data has an owner.

Every important data asset must have someone accountable for its meaning, quality, and appropriate use.

2. Quality is part of the product.

A dashboard, model, API, or AI system cannot be considered reliable when the underlying data is unreliable.

3. Access follows purpose.

People and systems receive access because they have a legitimate business need, not simply because access is technically convenient.

4. AI is accountable to people.

AI can assist with decisions, but accountability remains with the authorized business owner.

5. Automation does not remove responsibility.

A workflow executing automatically is still owned by the team that designed and operates it.

6. Security and privacy begin before deployment.

Data and AI controls must be considered during design rather than added only after a system is operational.

7. Measure business value.

Technical sophistication is not itself a business outcome.

8. Learn continuously.

Models, data, workflows, and business environments change. Production systems must therefore be monitored, reviewed, and improved throughout their useful life.